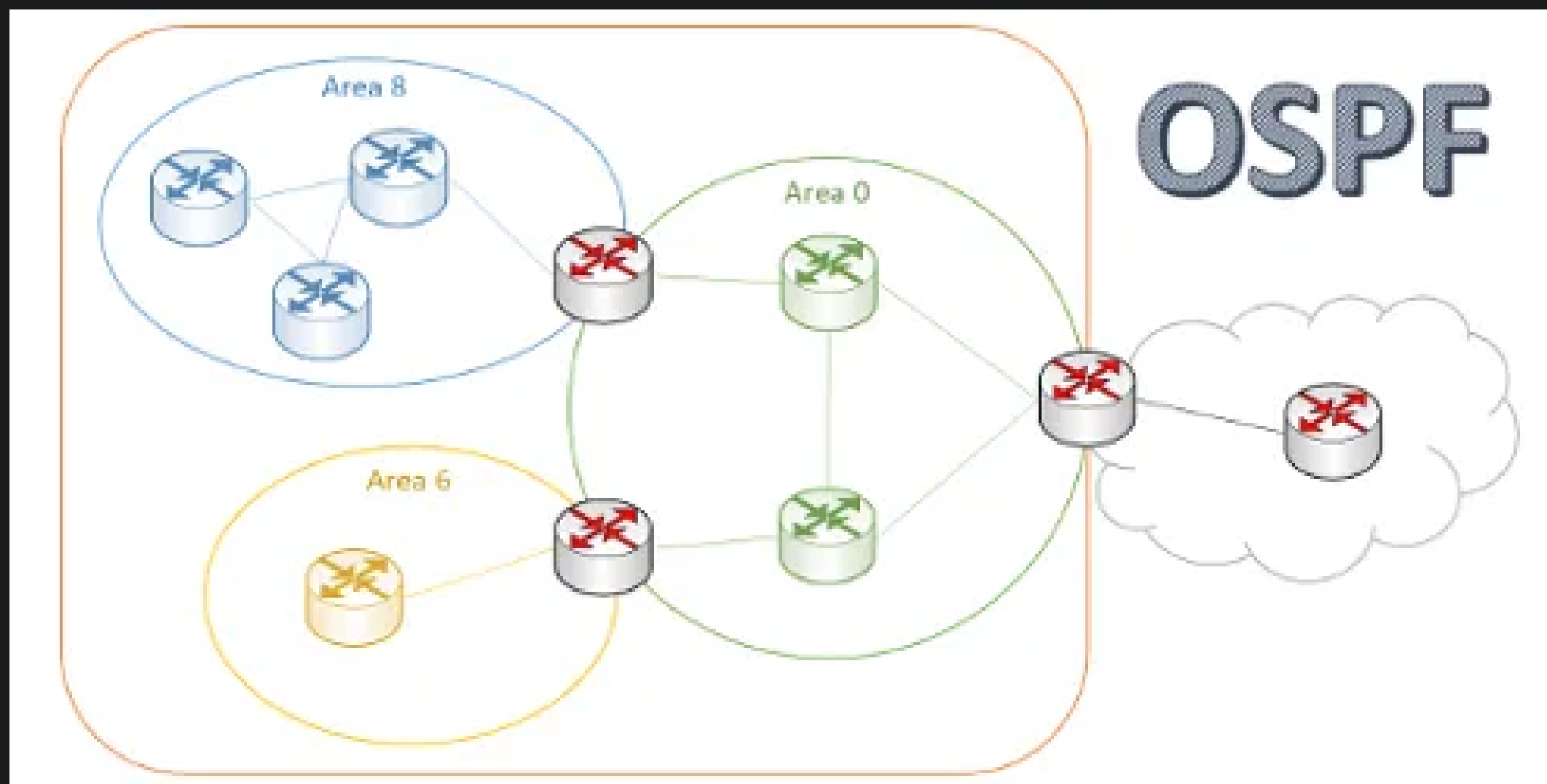


TP_ ROUTAGE DYNAMIQUE OSPF ET DÉCOUVERTE HSRP



DEFINITIONS

OSPF est un protocole de routage à état de lien qui vise à établir une table de routage optimale en utilisant l'algorithme SPF (Shortest Path First). Chaque routeur découvre ses voisins, échange des informations topologiques via des paquets spécifiques, et calcule les meilleures routes vers les sous-réseaux.



ETAPE 1 - VÉRIFICATIONS INITIALES (AVANT OSPF)

Vérifier connectivité postes → passerelle

```
PC-A
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.254

Pinging 192.168.0.254 with 32 bytes of data:

Reply from 192.168.0.254: bytes=32 time<1ms TTL=255
Reply from 192.168.0.254: bytes=32 time<1ms TTL=255
Reply from 192.168.0.254: bytes=32 time<1ms TTL=255
Reply from 192.168.0.254: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.0.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
PC-F2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.254

Pinging 192.168.0.254 with 32 bytes of data:

Reply from 192.168.0.254: bytes=32 time<1ms TTL=255
Reply from 192.168.0.254: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.0.254:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
```

```
PC-F1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.254

Pinging 192.168.0.254 with 32 bytes of data:

Reply from 192.168.0.254: bytes=32 time<1ms TTL=255
Reply from 192.168.0.254: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.0.254:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
```

Réponses OK
TTL 128 (PC → routeur)



VÉRIFICATIONS INITIALES (AVANT OSPF)

Vérifier connectivité entre routeurs voisins

```
Routeur-B#ping 20.5.5.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.5.5.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/34/53 ms
```

```
Routeur-B#ping 20.5.5.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.5.5.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/30/39 ms

Routeur-B#ping 20.6.6.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.6.6.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 30/39/47 ms

Routeur-B#ping 20.4.4.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.4.4.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 27/33/44 ms

Routeur-B#ping 20.2.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.2.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 35/40/54 ms
```

```
Routeur-C#ping 20.5.5.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.5.5.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 27/39/54 ms

Routeur-C#ping 20.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 17/24/33 ms
```

```
Routeur-D#ping 20.4.4.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.4.4.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/26/42 ms

Routeur-D#ping 20.3.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.3.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/38/59 ms
```

```
Routeur-E#ping 20.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 27/37/51 ms

Routeur-E#ping 20.1.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.1.1.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 37/43/57 ms

Routeur-E#ping 20.2.2.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.2.2.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 35/41/58 ms

Routeur-E#ping 20.3.3.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.3.3.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 27/40/60 ms
```

Tous les pings réussissent.



VÉRIFICATIONS INITIALES (AVANT OSPF)

Vérifier absence de routage avant OSPF

Depuis PC-A :

```
C:\>ping 172.12.0.254

Pinging 172.12.0.254 with 32 bytes of data:

Reply from 20.6.6.1: Destination host unreachable.
Reply from 20.6.6.1: Destination host unreachable.
Reply from 20.6.6.1: Destination host unreachable.
Reply from 20.6.6.1: Destination host unreachable.

Ping statistics for 172.12.0.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

→ réseau du site E

```
C:\>ping 20.0.0.1

Pinging 20.0.0.1 with 32 bytes of data:

Reply from 20.6.6.1: Destination host unreachable.
Reply from 20.6.6.1: Destination host unreachable.
Reply from 20.6.6.1: Destination host unreachable.
Reply from 20.6.6.1: Destination host unreachable.
```

→ réseau du site F

Les tests échouent et c'est complètement normal avant
OSPF



VÉRIFICATIONS INITIALES (AVANT OSPF)

Vérifier NAT sur RtA / RtF

```
Routeur-A>sh ip nat translations
Routeur-A>sh ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: Serial0/0/0
Inside Interfaces: FastEthernet0/1
Hits: 8 Misses: 28
Expired translations: 8
Dynamic mappings:
```

→ Routeur A

```
C:\>ping 8.8.8.8
```

```
Pinging 8.8.8.8 with 32 bytes of data:
```

```
Reply from 20.6.6.1: Destination host unreachable.
Reply from 20.6.6.1: Destination host unreachable.
Reply from 20.6.6.1: Destination host unreachable.
Reply from 20.6.6.1: Destination host unreachable.
```

→ PC A

```
Ping statistics for 8.8.8.8:
```

```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
Routeur-A>sh ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
icmp	20.6.6.2:21	192.168.0.1:21	8.8.8.8:21	8.8.8.8:21
icmp	20.6.6.2:22	192.168.0.1:22	8.8.8.8:22	8.8.8.8:22
icmp	20.6.6.2:23	192.168.0.1:23	8.8.8.8:23	8.8.8.8:23
icmp	20.6.6.2:24	192.168.0.1:24	8.8.8.8:24	8.8.8.8:24

→ Routeur A



ETAPE 2 - CONFIGURATION OSPF

Activer OSPF sur RtB, RtC, RtD, RtE

ROUTEUR RtB

```
RtB(config-if)#interface s0/0/0
RtB(config-if)#ip address 20.6.6.1 255.255.255.252
RtB(config-if)#no shutdown
RtB(config-if)#exit
RtB(config)#interface s0/0/1
RtB(config-if)#ip address 20.5.5.2 255.255.255.252
RtB(config-if)#no shutdown
RtB(config-if)#exit
RtB(config)#interface s0/1/0
RtB(config-if)#ip address 20.4.4.2 255.255.255.252
RtB(config-if)#no shutdown
RtB(config-if)#exit
RtB(config)#interface s0/1/1
RtB(config-if)#ip address 20.2.2.2 255.255.255.252
RtB(config-if)#no shutdown
RtB(config-if)#exit
```

```
RtB(config)#router ospf 1
RtB(config-router)#router-id 2.2.2.2
RtB(config-router)#network 20.6.6.0 0.0.0.3 area 0
RtB(config-router)#network 20.5.5.0 0.0.0.3 area 0
RtB(config-router)#network 20.4.4.0 0.0.0.3 area 0
RtB(config-router)#network 20.2.2.0 0.0.0.3 area 0
RtB(config-router)#end
```



ETAPE 2 - CONFIGURATION OSPF

Activer OSPF sur RtB, RtC, RtD, RtE

ROUTEUR RtC

```
Routeur-C#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Routeur-C(config)#hostname RtC
RtC(config)#interface s0/0/0
RtC(config-if)#ip address 20.1.1.2 255.255.255.252
RtC(config-if)#no shutdown
RtC(config-if)#exit
RtC(config)#interface s0/0/1
RtC(config-if)#ip address 20.5.5.1 255.255.255.252
RtC(config-if)#no shutdown
RtC(config-if)#exit
RtC(config)#router ospf 1
RtC(config-router)#router-id 3.3.3.3
RtC(config-router)#network 20.1.1.0 0.0.0.3 area 0
RtC(config-router)#network 20.5.5.0 0.0.0.3 area 0
RtC(config-router)#end
```



ETAPE 2 - CONFIGURATION OSPF

Activer OSPF sur RtB, RtC, RtD, RtE

ROUTEUR RtD

```
Routeur-D>enable
Routeur-D#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Routeur-D(config)#hostname RtD
RtD(config)#interface s0/1/0
RtD(config-if)#ip address 20.4.4.1 255.255.255.252
RtD(config-if)#no shutdown
RtD(config-if)#exit
RtD(config)#interface s0/0/1
RtD(config-if)#ip address 20.3.3.2 255.255.255.252
RtD(config-if)#no shutdown
RtD(config-if)#exit
RtD(config)#interface s0/0/0
RtD(config-if)#ip address 20.7.7.1 255.255.255.252
RtD(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
RtD(config-if)#exit
RtD(config)#router ospf 1
RtD(config-router)#router-id 4.4.4.4
RtD(config-router)#network 20.4.4.0 0.0.0.3 area 0
RtD(config-router)#network 20.3.3.0 0.0.0.3 area 0
RtD(config-router)#network 20.3.3.0 0.0.0.3 area 0
00:39:01: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial0/1/0 from
00:39:06: %OSPF-5-ADJCHG: Process 1, Nbr 5.5.5.5 on Serial0/0/1 from LOADING to FULL,
Loading Done

RtD(config-router)#network 20.7.7.0 0.0.0.3 area 0
RtD(config-router)#end
RtD#
%SYS-5-CONFIG I: Configured from console by console
```

ETAPE 2 - CONFIGURATION OSPF

Activer OSPF sur RtB, RtC, RtD, RtE

ROUTEUR RtE

```
Routeur-E#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Routeur-E(config)#hostname RtE
RtE(config)#interface s0/0/0
RtE(config-if)#ip address 20.0.0.2 255.255.255.252
RtE(config-if)#no shutdown
RtE(config-if)#exit
RtE(config)#interface s0/0/1
RtE(config-if)#ip address 20.3.3.1 255.255.255.252
RtE(config-if)#no shutdown
RtE(config-if)#exit
RtE(config)#interface s0/1/0
RtE(config-if)#ip address 20.1.1.1 255.255.255.252
RtE(config-if)#no shutdown
RtE(config-if)#exit
RtE(config)#interface s0/1/1
RtE(config-if)#ip address 20.2.2.1 255.255.255.252
RtE(config-if)#no shutdown
RtE(config-if)#exit
RtE(config)#router ospf 1
RtE(config-router)#router-id 5.5.5.5
RtE(config-router)#network 20.0.0.0 0.0.0.3 area 0
RtE(config-router)#network 20.3.3.0 0.0.0.3 area 0
RtE(config-router)#network 20.1.1.0 0.0.0.3 area 0
RtE(config-router)#
02:42:33: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on Serial0/1/0 from LOADING to FULL, Loading
Done

RtE(config-router)#network 20.2.2.0 0.0.0.3 area 0
RtE(config-router)#
02:42:51: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial0/1/1 from LOADING to FULL, Loading
Done

RtE(config-router)#end
```

ETAPE 2 - CONFIGURATION OSPF

Activer OSPF sur RtB, RtC, RtD, RtE

ROUTEUR RtE

```
RtE>enable
RtE#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
RtE(config)#hostname RtE
RtE(config)#interface s0/0/0
RtE(config-if)#ip address 20.0.0.2 255.255.255.252
RtE(config-if)#no shutdown
RtE(config-if)#exit
RtE(config)#interface s0/0/1
RtE(config-if)#ip address 20.3.3.1 255.255.255.252
RtE(config-if)#no shutdown
RtE(config-if)#exit
RtE(config)#interface s0/1/0
RtE(config-if)#ip address 20.1.1.1 255.255.255.252
RtE(config-if)#no shutdown
RtE(config-if)#exit
RtE(config)#interface s0/1/1
RtE(config-if)#ip address 20.2.2.1 255.255.255.252
RtE(config-if)#no shutdown
RtE(config-if)#exit
RtE(config)#router ospf 1
RtE(config-router)#router-id 5.5.5.5
RtE(config-router)#network 20.0.0.0 0.0.0.3 area 0
RtE(config-router)#network 20.3.3.0 0.0.0.3 area 0
RtE(config-router)#network 20.1.1.0 0.0.0.3 area 0
RtE(config-router)#network 20.2.2.0 0.0.0.3 area 0
RtE(config-router)#end
RtE#
%SYS-5-CONFIG_I: Configured from console by console
```

ÉTAPE 3 – TESTS OSPF (BON FONCTIONNEMENT)

Vérifier voisinage OSPF:

```
RtB>enable
RtB#show ip ospf neighbor
```

Neighbor ID	Pri	State		Dead Time	Address	Interface
3.3.3.3	0	FULL/	-	00:00:33	20.5.5.1	Serial0/0/1
4.4.4.4	0	FULL/	-	00:00:37	20.4.4.1	Serial0/1/0
5.5.5.5	0	FULL/	-	00:00:37	20.2.2.1	Serial0/1/1

Voisin FULL avec RtA ? (non — RtA n’est pas OSPF)
Voisin FULL avec RtC
Voisin FULL avec RtD
Voisin FULL avec RtE

Vérifier routes OSPF:

```
RtB#show ip route ospf
```

```
      20.0.0.0/8 is variably subnetted, 11 subnets, 2 masks
O       20.0.0.0 [110/128] via 20.2.2.1, 00:44:55, Serial0/1/1
O       20.1.1.0 [110/128] via 20.5.5.1, 00:44:55, Serial0/0/1
        [110/128] via 20.2.2.1, 00:44:55, Serial0/1/1
O       20.3.3.0 [110/128] via 20.4.4.1, 00:06:14, Serial0/1/0
        [110/128] via 20.2.2.1, 00:06:14, Serial0/1/1
```

Les lignes doivent commencer par un “ O ”
O est une route apprise automatiquement par OSPF
Le routeur n’a pas configuré cette route manuellement.
Il l’a reçue d’un voisin OSPF (RtB, RtC, RtD...).



ÉTAPE 3 – TESTS OSPF (BON FONCTIONNEMENT)

Test ping à travers le réseau: Tous les pings doivent réussir.

```
C:\>ping 172.12.0.254

Pinging 172.12.0.254 with 32 bytes of data:

Reply from 172.12.0.254: bytes=32 time=14ms TTL=253
Reply from 172.12.0.254: bytes=32 time=15ms TTL=253

Ping statistics for 172.12.0.254:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 15ms, Average = 14ms

Control-C
^C
C:\>ping 192.168.0.254

Pinging 192.168.0.254 with 32 bytes of data:

Reply from 192.168.0.254: bytes=32 time<1ms TTL=255
Reply from 192.168.0.254: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.0.254:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Traceroute:

```
C:\>tracert 172.12.0.254

Tracing route to 172.12.0.254 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms   192.168.0.254
  1  6 ms    6 ms    6 ms   20.6.6.1
  2  6 ms    0 ms    4 ms   172.12.0.254

Trace complete.
```

RtA → RtB → RtE → destination



ÉTAPE 4 – MODE DÉGRADÉ (RUPTURE LIEN OSPF)

Couper un lien: Sur RtB, coupe S0/1/1 (vers RtE)

```
RtB#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
RtB(config)#interface s0/1/1
RtB(config-if)#shutdown

RtB(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to down

02:54:01: %OSPF-5-ADJCHG: Process 1, Nbr 5.5.5.5 on Serial0/1/1 from FULL to DOWN,
Neighbor Down: Interface down or detached
```

Observer le voisinage après la coupure: le voisin 5.5.5.5 doit disparaître

```
RtB#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	0	FULL/ -	00:00:32	20.5.5.1	Serial0/0/1
4.4.4.4	0	FULL/ -	00:00:36	20.4.4.1	Serial0/1/0



ÉTAPE 4 – MODE DÉGRADÉ (RUPTURE LIEN OSPF)

Vérifier la nouvelle route

Avant coupure on avait: O 20.0.0.0/30 via 20.2.2.1 ← route directe

```
RtB#show ip route ospf
      20.0.0.0/8 is variably subnetted, 11 subnets, 2 masks
O       20.0.0.0 [110/128] via 20.2.2.1, 00:44:55, Serial0/1/1
O       20.1.1.0 [110/128] via 20.5.5.1, 00:44:55, Serial0/0/1
        [110/128] via 20.2.2.1, 00:44:55, Serial0/1/1
O       20.3.3.0 [110/128] via 20.4.4.1, 00:06:14, Serial0/1/0
        [110/128] via 20.2.2.1, 00:06:14, Serial0/1/1
```

Après coupure:

```
RtB#show ip route ospf
      20.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
O       20.0.0.0 [110/192] via 20.5.5.1, 00:03:26, Serial0/0/1
        [110/192] via 20.4.4.1, 00:03:26, Serial0/1/0
O       20.1.1.0 [110/128] via 20.5.5.1, 00:03:26, Serial0/0/1
O       20.3.3.0 [110/128] via 20.4.4.1, 00:03:26, Serial0/1/0
      172.12.0.0/24 is subnetted, 1 subnets
O       172.12.0.0 [110/129] via 20.5.5.1, 00:03:26, Serial0/0/1
        [110/129] via 20.4.4.1, 00:03:26, Serial0/1/0
```

← route alternative par RtC



ÉTAPE 4 – MODE DÉGRADÉ (RUPTURE LIEN OSPF)

Traceroute avant la coupure

```
C:\>tracert 172.12.0.254

Tracing route to 172.12.0.254 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.0.254
  2    6 ms      6 ms      6 ms      20.6.6.1
  3    6 ms      0 ms      4 ms      172.12.0.254

Trace complete.
```

Traceroute après la coupure

```
C:\>tracert 172.12.0.254

Tracing route to 172.12.0.254 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.0.254
  2    1 ms      2 ms      6 ms      20.6.6.1
  3    6 ms      8 ms      2 ms      20.5.5.1
  4    5 ms      6 ms      6 ms      172.12.0.254
```

→ Les sauts changent :

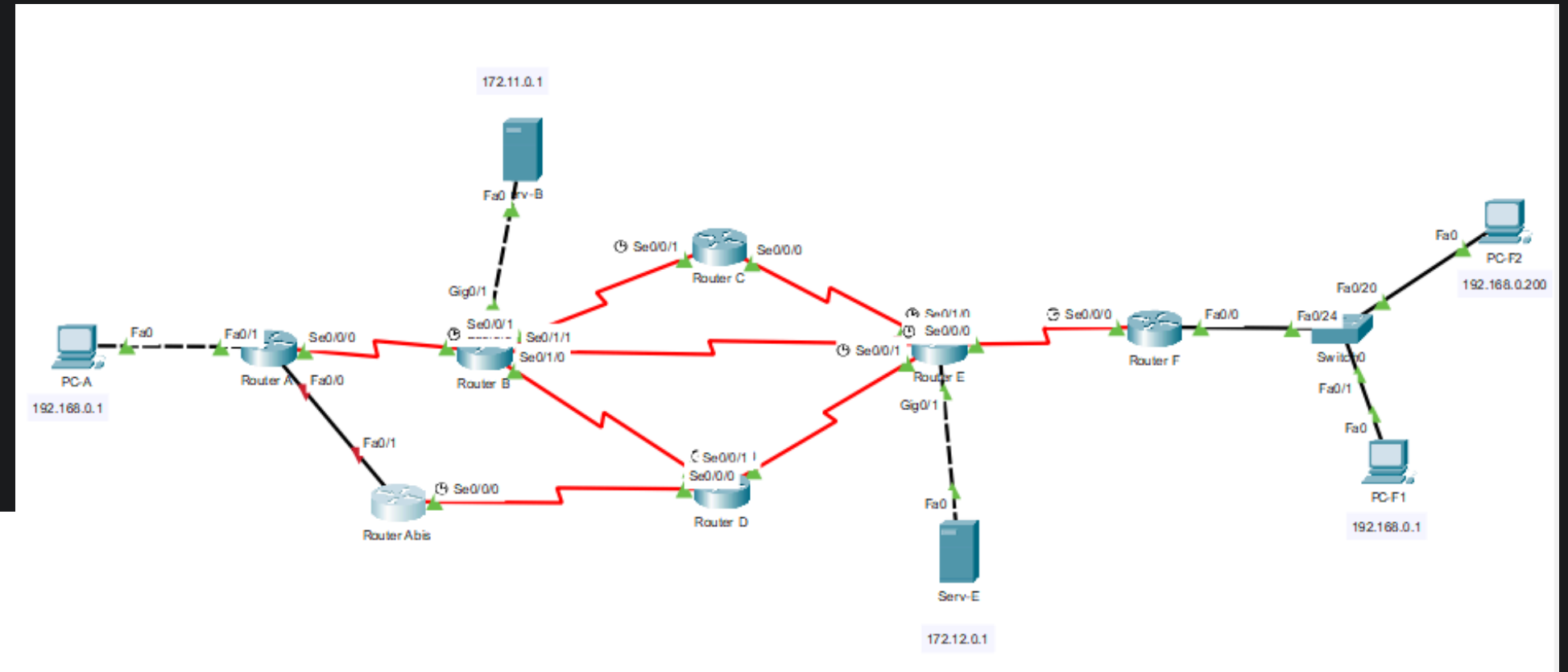
Avant : RtA → RtB → RtE

Après : RtA → RtB → RtC → RtE
(mode dégradé)



ÉTAPE 5 – CONFIGURATION HSRP (REDONDANCE SITE A)

```
Routeur-Abis(config)#conf t
%Invalid hex value
Routeur-Abis(config)#hostname RtA-bis
RtA-bis(config)#interface fa0/1
RtA-bis(config-if)#ip address 192.168.0.253 255.255.255.0
RtA-bis(config-if)#no shutdown
RtA-bis(config-if)#exit
RtA-bis(config)#interface s0/0/0
RtA-bis(config-if)#ip address 20.7.7.2 255.255.255.252
RtA-bis(config-if)#no shutdown
RtA-bis(config-if)#exit
RtA-bis(config)#router ospf 1
RtA-bis(config-router)#router-id 7.7.7.7
RtA-bis(config-router)#network 20.7.7.0 0.0.0.3 area 0
RtA-bis(config-router)#
00:09:24: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on Serial0/0/0 from LOADING to
Loading Done
end
RtA-bis#
%SYS-5-CONFIG_I: Configured from console by console
```



ÉTAPE 5 – CONFIGURATION HSRP (REDONDANCE SITE A)

Config RtA

```
Routeur-A(config-if)#interface fa0/0
Routeur-A(config-if)#standby 1 ip 192.168.0.254
% Warning: address is not within a subnet on this interface
Routeur-A(config-if)#standby 1 priority 150
Routeur-A(config-if)#standby 1 preempt
Routeur-A(config-if)#
```

Config RtA-bis

```
RtA-bis(config)#interface fa0/1
RtA-bis(config-if)#ip address 192.168.0.253 255.255.255.0
RtA-bis(config-if)#no shutdown
RtA-bis(config-if)#standby 1 ip 192.168.0.254
RtA-bis(config-if)#standby 1 priority 100
RtA-bis(config-if)#standby 1 preempt
RtA-bis(config-if)#end
```

